

Regular Set

Regular Expressions. - Syntax to define regular sets.

Finite state Automata - Computational model for processing regular sets.

Syntax specification for a PL.

$\langle \text{if-stmt} \rangle ::= \text{if } \langle \text{bool-exp} \rangle \text{ then } \langle \text{statement} \rangle$
 $\text{else } \langle \text{statement} \rangle$

$\langle \text{arith-exp} \rangle ::= \langle \text{var} \rangle \mid \langle \text{const} \rangle \mid$

$\langle \text{arith-exp} \rangle \langle \text{arith-op} \rangle \langle \text{arith-exp} \rangle$

$\langle \text{arith-op} \rangle ::= + \mid - \mid * \mid /$

$\langle \text{const} \rangle ::= 0 \mid 1 \mid 2 \mid \dots \mid 8 \mid 9$

$\langle \text{var} \rangle ::= a \mid b \mid c \mid \dots \mid x \mid y \mid z$

$A = \{a^n b^n \mid n \geq 0\}$. - not regular.

Recursive definition

- $\epsilon \in A$
- if $w \in A$ then $awb \in A$.

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Context Free Languages.

Context Free Grammars.

$G = (N, \Sigma, P, S)$.

N - finite set (non-terminal symbols).

Σ - finite set (terminal symbols)

Assumption : $N \cap \Sigma = \emptyset$.

P - finite subset of $N \times (N \cup \Sigma)^*$ [productions]
 $A \rightarrow \alpha$

$S \in N$ (start symbol).

Notation . A, B, C - non terminal symbols
 a, b, c - terminal symbols

α, β, γ - strings over $(N \cup \Sigma)^*$

$Z = \{a^n b^n \mid n \geq 0\}$ - is a CFL.

CFG. $G = (N, \Sigma, P, S)$ $N = \{S\}$, $\Sigma = \{a, b\}$.

$P = \{S \rightarrow aSb, S \rightarrow \epsilon\}$.

$S \rightarrow aSb \mid \epsilon$

$a^3 b^3$: $S \xrightarrow[G]{1} aSb \xrightarrow[G]{1} aaSbb$

$\xrightarrow[G]{1} aaaSbbb \xrightarrow[G]{1} aaabbbb.$

Context Free Grammars.

$$G = (N, \Sigma, P, S).$$

N - finite set (non-terminal symbols).

Σ - finite set (terminal symbols). $\Sigma \cap N = \emptyset$

P - finite subset of $N \times (N \cup \Sigma)^*$ [productions]

$$P \subseteq N \times (N \cup \Sigma)^* \quad \{(A, \alpha_1), (A, \alpha_2), (A, \alpha_3)\} \subseteq P$$

$$A \rightarrow \alpha_1 \mid \alpha_2 \mid \alpha_3.$$

$S \in N$ (start symbol).

Suppose $\alpha, \beta \in (N \cup \Sigma)^*$. β is derivable from α

in one step $\alpha \xrightarrow{1} \beta$ if β can be obtained

from α by replacing some occurrence of a non-terminal A in α with γ where $A \rightarrow \gamma \in P$.

if $\exists \alpha_1, \alpha_2 \in (N \cup \Sigma)^*$ s.t. $\alpha = \alpha_1 A \alpha_2$ and

$\exists A \rightarrow \gamma \in P$ then $\alpha \xrightarrow[G]{1} \beta = \alpha_1 \gamma \alpha_2$

$\alpha \xrightarrow[G]{1} \beta$ - one step derivation

$\xrightarrow[G]{*}$: reflexive transitive closure of the relation $\xrightarrow[G]{1}$.

$\alpha \xrightarrow[G]{0} \alpha$ for all α

$\alpha \xrightarrow[G]{n+1} \beta$ if $\exists \gamma$ s.t. $\alpha \xrightarrow[G]{n} \gamma$ and $\gamma \xrightarrow[G]{1} \beta$.

$\alpha \xrightarrow[G]{*} \beta$ if $\alpha \xrightarrow[G]{n} \beta$ for some $n \geq 0$.

Language generated by G .

$$L(G) = \{x \in \Sigma^* \mid S \xrightarrow[G]{*} x\}$$

$B \subseteq \Sigma^*$ is a context free language (CFL) if

$B = L(G)$ for some CFG G .

$Z = \{a^n b^n \mid n \geq 0\}$ - is a CFL.

CFG. $G = (N, \Sigma, P, S)$ $N = \{S\}$, $\Sigma = \{a, b\}$.

$P = \{S \rightarrow aSb, S \rightarrow \epsilon\}$.

$L(G) = Z$.

$a^3 b^3$: $S \xrightarrow[G]{1} aSb \xrightarrow[G]{1} a aSb b$
 $\xrightarrow[G]{1} a a aSb b b \xrightarrow[G]{1} a a a b b b$.

By induction on n we can show: $S \xrightarrow[G]{n+1} a^n b^n$

$\Rightarrow$ all strings of the form $a^n b^n \in L(G)$.

Conversely, the only strings in $L(G)$ are of the form

$a^n b^n$ - induction on the length of the derivation.